

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
GENERAL CERTIFICATE OF EDUCATION ADVANCED LEVEL

CHEMISTRY
PAPER 2

6031/2
1 hour 30 minutes

JUNE 2026 SESSION

Candidates answer on the question paper.

Additional materials:

Data Booklet

Electronic Calculator B6aeQ

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question

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1	
2	
3	
4	
5	
6	
TOTAL	

This question paper consists of 13 printed pages.

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Answer all questions

Write your answers in the spaces provided on the question paper.

- 1 (a) Fig 1.1 shows successive $\log_{10}$ ionisation energies of an unknown element Q.

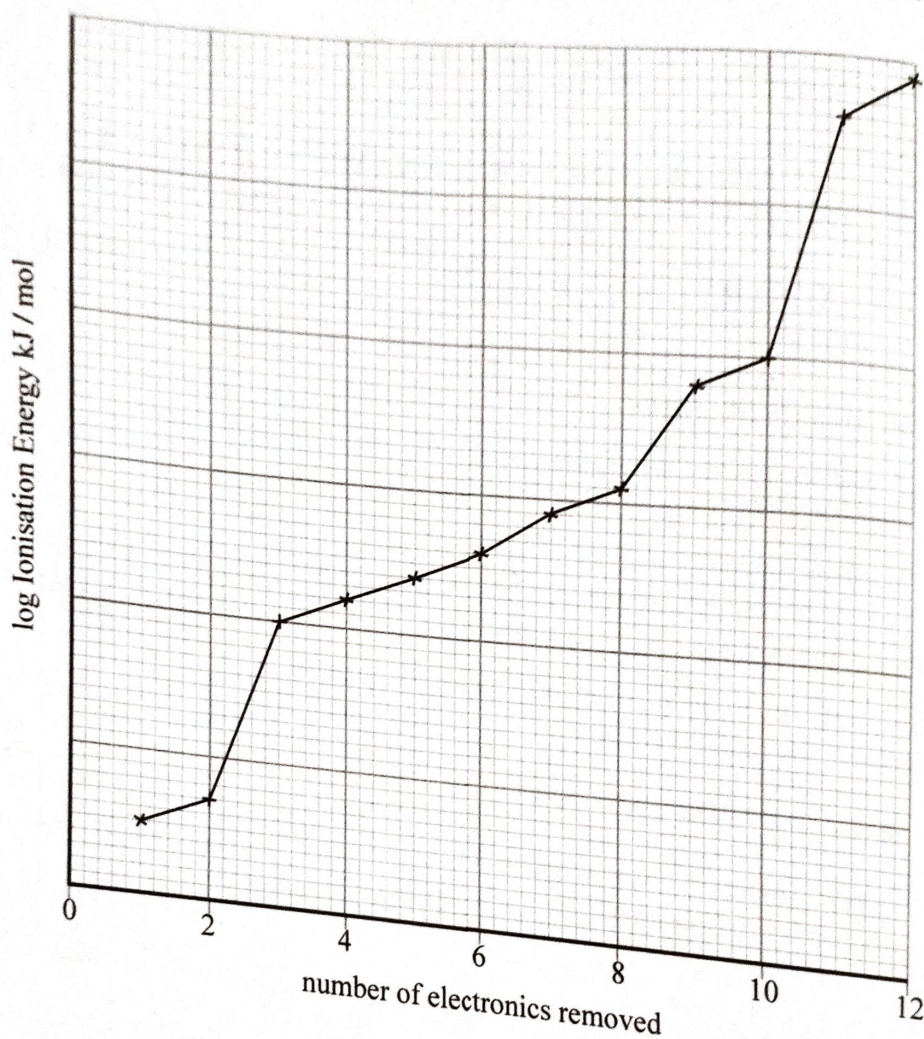


Fig 1.1

- (i) Deduce, with a reason, the group to which element Q belongs.

- (ii) Write down the electronic configuration of Q.

[2]

[1]



- (iii) Explain why the ionisation energy of Mg^{2+} is greater than that of Ca^{2+} .

[2]

- (b) Explain why molten MgCl_2 conducts electricity.

[1]

- (c) (i) A volume of 20.00 cm^3 of a gaseous hydrocarbon was completely burnt in 90.00 cm^3 of oxygen to give 60.00 cm^3 carbon dioxide. Deduce the molecular formula of the hydrocarbon.

[2]

- (ii) Draw all the possible isomers of the hydrocarbon.

[2]



- 2 (a) A sub atomic particle passed through an electric field without deflection.

(i) Name the sub atomic particle.

_____ [1]

(ii) Fig 2.1 shows two subatomic particles passing through an electric field.

Complete Fig 2.1 to show the effect of the electric field on particles.

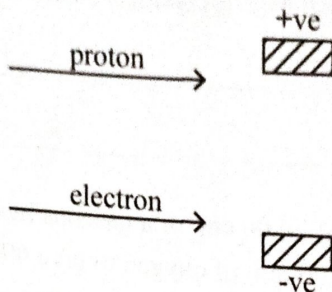


Fig 2.1

- (b) When 0.100 g of ammonium nitrate is heated in excess aqueous sodium hydroxide, 8.88cm^3 of 0.100mol dm^{-3} HCl is needed to neutralise the gas liberated.

(i) Write a balanced ionic equation to show the chemical change that occurs.

(ii) Describe a chemical test to identify the gas evolved.



- (iii) Calculate the number of moles of ammonia gas liberated.

_____ [1]

- (iv) Deduce the percentage of ammonium nitrate in the sample.

_____ [2]

- (c) State any **two** assumptions made for real gases to behave ideally

1. _____

2. _____

_____ [2]



- 3 (a) (i) Sketch a Boltzmann distribution curve to illustrate the effect of vanadium (V) oxide in the Contact process

[1]

- (ii) State and explain the optimum conditions employed in the Contact process.

[4]

- (b) Fig 2.2 shows a reaction scheme for the production of an aqueous solution of calcium chloride.

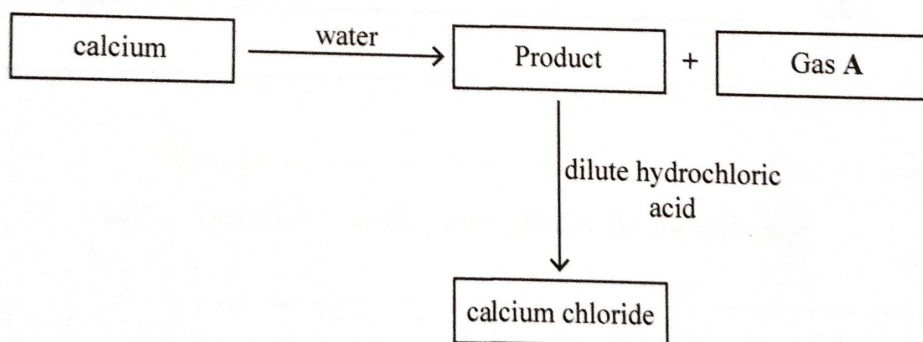


Fig 2.2

- (i) Identify gas A.

[1]



- (ii) Describe the test for gas A.

_____ [1]

- (iii) Explain why CaCl_2 is preferred as a drying agent for acidic gases to CaO .

_____ [1]

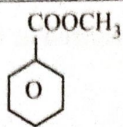
- (c) Compare thermal stabilities of barium carbonate and magnesium carbonate.

_____ [2]



- 4 (a) Table 4.1 shows the formula of the organic compounds **W** and **Z**.

Table 4.1

W	$\text{C}(\text{CH}_3)_2\text{CHCH}_2\text{CH}(\text{CH}_3)_2$
Z	

- (i) Name compound

1. **W**, _____
2. **Z**, _____ [2]

- (ii) State the homologous series to which **W** belongs.

_____ [1]

- (b) Fig 4.1 shows how nitrobenzene can be converted to other organic compounds, **A**, **B** and **C**.

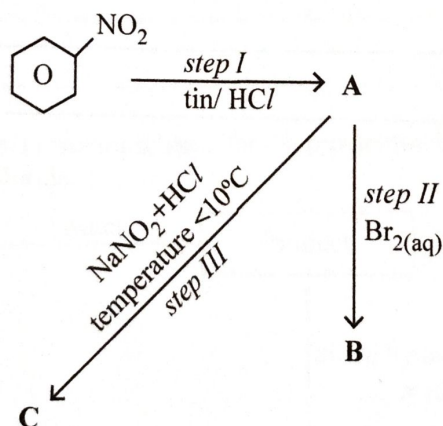


Fig 4.1

- (i) Give the structural formula of

1. **A**, _____
2. **B**, _____

[2]



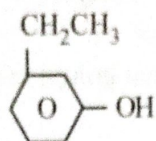
- (ii) Name the type of reaction in step I.

_____ [1]

- (iii) Explain why the temperature must be kept below 10°C in step III.

_____ [1]

- (iv) C reacts with the organic compound shown in Fig. 4.2.



Draw the displayed formula of the organic product formed.

[1]

- (c) Outline the preparation of ethanal from ethene.

_____ [2]

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- 5 (a) Fig 5.1 shows the displayed structural formula of an organic compound, estrone.

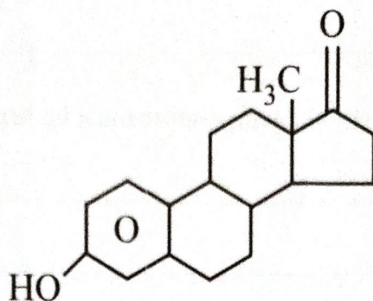


Fig 5.1

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- (i) Name the functional groups in the compound.

_____ [2]

- (ii) Describe a chemical test to identify each of the functional groups in the compound.

_____ [2]



(iii) Draw the structure of an organic product formed when the compound reacts with

1. aqueous bromine

2. LiAlH_4 in dry ether

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[2]

(b) Fig 5.2 shows how an organic compound A can be converted to other organic compounds.

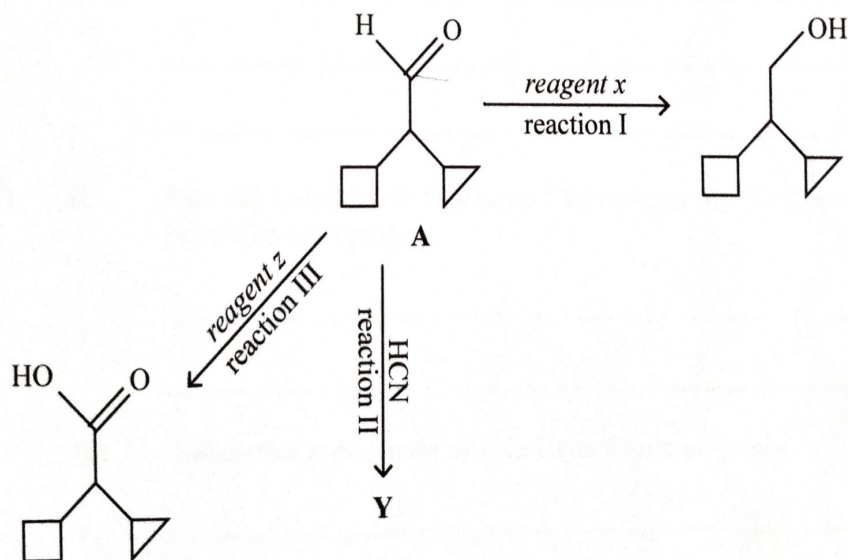


Fig. 5.2

(i) Name the type of reaction in

1. I,

2. III,

[2]

(ii) Draw a skeletal formula of product Y.

[1]

(iii) Identify reagent Z.

[1]



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(a)

(i)

Distinguish between a transition element and a non transition element.

[1]

(ii)

Describe what happens when excess ammonia solution is added to aqueous copper (II) ions.

[2]

(iii)

Explain the origin of the green colour in Fe^{2+} compounds.

[4]

(b)

(i)

State the advantage of Thin Layer Chromatography (TLC) over Paper Chromatography.

[1]

(ii)

Outline **two** applications of Thin Layer Chromatography.

[2]



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